

End-to-End Business Analytics Case Study

Business Performance Dashboard & Analysis

Objective

Clean and explore the supplied business dataset in Python, analyze it using Pandas/SQL logic, visualize the results in Excel and a BI tool, and communicate stakeholder-ready insights and recommendations.

1. Data Preparation

The workbook contains 700 records and 16 original fields. Date fields were standardized to datetime, column names were trimmed, and 53 missing Discount Band values were labeled 'Unknown' so they were retained rather than silently dropped. Duplicate rows were checked/removed in the Python workflow. A derived Sales Margin % metric was created as Profit / Sales.

2. Executive KPI Snapshot

Metric	Value
Sales	\$118,726,350.26
Profit	\$16,893,702.26
Profit Margin	14.23%
Units Sold	1,125,806
Discounts	\$9,205,248.24
COGS	\$101,832,648.00

3. Key Findings

1. Government is the largest sales segment at \$52,504,260.67 and generates \$11,388,173.17 profit.
2. Enterprise records an aggregate loss of \$-614,545.62, despite meaningful sales volume. This is the clearest segment-level profitability issue to investigate.
3. Channel Partners has the highest segment margin (73.13%), but its sales scale is much smaller than Government and Small Business.
4. The High discount band has a 9.07% margin versus 17.87% for Low discounts. This suggests discount intensity should be monitored alongside profitability.
5. Paseo is the top product by sales (\$33,011,143.95) and generates \$4,797,437.95 profit.
6. France has the highest country-level margin (15.66%) among the five countries in the dataset.
7. Monthly sales are uneven, with strong peaks in October 2014 (\$12,375,819.92) and December 2014 (\$11,998,787.90).

4. Recommendations

- Investigate Enterprise-level cost structure, pricing, and discounting to identify the drivers of negative profit.
- Review discount policies using both sales growth and margin impact rather than sales alone.
- Protect and expand high-margin segments/products where scale allows.
- Use monthly performance monitoring to plan inventory, sales capacity, and promotions around high/low periods.

- In the BI dashboard, allow filtering by year, country, segment, product, and discount band so stakeholders can drill into the drivers.

5. Dashboard Design

The Excel dashboard includes KPI cards, segment sales/profit, monthly sales trend, product profit, and country sales. The Power BI package includes cleaned data, DAX measures, slicer recommendations, and a one-page dashboard build specification.

6. Limitations

The dataset is observational. Segment, discount, product, and country differences are descriptive associations and should not be treated as causal effects without further testing. The source data also contains missing Discount Band values, which are explicitly retained as Unknown.